

Newton’ s cell action as a measure on cuts

After refereeing (GPT-6 Astra, 2026-09-28). A6: ACCEPT Proposition 6’ s Cameron–Martin norm, optimal equal-prior error and exact constant in the Planck mesh. RE-FINE the statement to two prescribed Gaussian bridge means with the same pinned endpoints and covariance, distinguish the variance rate from the diffusion coefficient, and identify the likelihood pairing as a Gaussian stochastic functional. The agreement with the mark model is structural: both have the sharp squared signal distance $2K_\tau/\hbar$ from the same Dirichlet energy. The bridge test attains its bound; the mark model approaches it by dense protocols. Both supply $\hbar > 0$, leaving necessity unchanged.

After refereeing (Fable, 2026-09-28). Theorems 1–2, Corollary 3’ s arithmetic and the quotations were re-derived and accepted. Corrections applied: Proposition 4 now names the Kullback–Leibler divergence, the shift δx , the phase K_τ/ε and the Cameron–Martin normalizer, and its link to Rivero 1998 is stated as a counterpart; Corollary 5 separates a floor on a cell’ s action from a floor on a cut’ s share (a factor $\frac{3}{4}$ for halving); § 3 labels the two-quarter-cut schedule a construction and adds the alternating reading favoured by 斲必半; the Lévy–Ciesielski comparison names the right quantity.

Result, 2026-09-27 (atlas § 1b, user remarks on arbitrary cuts and on the rod). Let a constant force F act on a body of mass m over a cell of duration τ , and let δx be the chord minus the parabola (the chord is the inertial comparison path with the same endpoints). Its Galileo action

$$K_\tau = \frac{m}{2} \int_0^\tau \dot{\delta x}^2 dt = \frac{F^2 \tau^3}{24m}$$

(the two-path note writes the same quantity as $(F/4v)A = \tau \Delta E/12$) behaves as a measure on the process of cutting the cell.

- **Theorem 1 (one cut).** A cut at fraction $s \in (0, 1)$ spends exactly $3s(1-s)K_\tau$, and this share equals the Cameron–Martin energy $\frac{m}{2} \int \dot{\phi}^2$ of the Schauder hat ϕ that the cut inserts, of height the sagitta $Fs(1-s)\tau^2/(2m)$.
- **Theorem 2 (any cut sequence).** The hats of successive cuts are orthogonal in the energy $\int \dot{f} \dot{g}$, so the spent shares add. Along any sequence of cuts, at any positions and in any order, the total spent equals $K_\tau - \sum_i K_{\tau_i}$ over the current pieces, and it tends to K_τ iff the mesh $\max_i \tau_i$ tends to zero; the remainder is at most $K_\tau \max_i \tau_i^2/\tau^2$.
- **Corollary 3 (the rod and the Mohist schedules).** Halving only the remaining piece each day, the schedule of the stick in Zhuangzi 33, spends $\frac{6}{7}K_\tau$, and so does one halving a day taken alternately from the front and the back. Removing a quarter from each end each day (two quarter-cuts a day, a construction motivated by the Mohist “taking from front and back”) spends $\frac{27}{28}K_\tau$. All three leave a definite share unspent forever, because they keep cut-off pieces of fixed length.
- **Proposition 4 (weight of a cut).** Under the Euclidean path measure of a free particle with variance rate $\hbar/m$ (Wiener measure), the force signal in one cut has Kullback–Leibler divergence exactly (spent share)/ $\hbar$. In real time the same share, divided by the action resolution, is the phase of the two-path comparison.
- **Proposition 6 (the mark mesh as a Cameron–Martin threshold; 2026-09-28, refereed ACCEPT with the hypotheses of § 5b).** Under the same path measure, the optimal equal-prior test between force F and no force in a cell, observing the whole path with pinned ends, errs with probability $\Phi(-\sqrt{K_\tau/2\hbar})$. It reaches ϵ iff $K_\tau \geq 2z_{1-\epsilon}^2 \hbar$, which is exactly the Planck paper’ s mark mesh $\tau_* = (48z_{1-\epsilon}^2 m \hbar / F^2)^{1/3}$, obtained there from real-time Gaussian marks with Robertson’ s bound.

- **Corollary 5 (a floor counts the cuts).** If every exhibited cut must spend at least an action $\kappa > 0$, at most K_τ/κ cuts can be exhibited, whatever the schedule. For halving refinement, level n is exhibited iff $\frac{3}{4}K_\tau 8^{-n} \geq \kappa$. The Planck paper's mark mesh $\tau_* = (48z_{1-\epsilon}^2 m\hbar/F^2)^{1/3}$ is the cell whose action is $2z_{1-\epsilon}^2 \hbar$; its own halving spends $\frac{3}{2}z_{1-\epsilon}^2 \hbar$.

Theorems 1 and 2 are elementary, and their content lies in the reading they give: Newton's refinement is a process that spends a fixed, finite action, with the nested additivity of the Lévy–Ciesielski construction of the Brownian bridge. Proposition 4 makes the correspondence exact: the action is the Cameron–Martin energy of the force's shift, decomposed over the hats. Corollary 5 is a consistency statement. It says where a floor, if present, stops the cutting, and leaves the necessity question of the fifth-postulate note where it was.

1. One cut

Put $C = F^2/(24m)$, so $K_\tau = C\tau^3$. On $[0, \tau]$ the parabola with acceleration F/m and the chord through its endpoints differ by

$$\delta x(t) = \frac{F}{2m} t(\tau - t), \quad \dot{\delta x}(t) = \frac{F}{m} \left(\frac{\tau}{2} - t \right),$$

$$\text{so } \frac{m}{2} \int_0^\tau \dot{\delta x}^2 dt = \frac{m}{2} \cdot \frac{F^2}{m^2} \cdot \frac{\tau^3}{12} = C\tau^3.$$

Cut at $t_c = s\tau$. The polygon through the three points $0, s\tau, \tau$ differs from the chord by the hat ϕ of height $h = \delta x(s\tau) = Fs(1-s)\tau^2/(2m)$, linear on $[0, s\tau]$ and on $[s\tau, \tau]$. The polygon differs from the parabola, on each piece, by the chord-minus-parabola of that piece: $\delta x = \phi + \delta x_1 + \delta x_2$ with δx_j supported on piece j and vanishing at its ends.

Orthogonality. $\dot{\phi}$ is constant on each piece and $\int_{\text{piece}} \dot{\delta x}_j = 0$, so $\int \dot{\phi} \dot{\delta x}_j = 0$; the δx_j have disjoint supports. Hence

$$K_\tau = \frac{m}{2} \int \dot{\phi}^2 + K_{s\tau} + K_{(1-s)\tau}.$$

The hat's energy. $\frac{m}{2} \int \dot{\phi}^2 = \frac{m}{2} h^2 \left(\frac{1}{s\tau} + \frac{1}{(1-s)\tau} \right) = \frac{mh^2}{2s(1-s)\tau} = \frac{F^2 s(1-s)\tau^3}{8m} = 3s(1-s)C\tau^3$. The same number follows from $\tau^3 = (s\tau)^3 + ((1-s)\tau)^3 + 3s(1-s)\tau^3$, which is the algebraic form of Theorem 1. The share is largest for halving, $\frac{3}{4}K_\tau$. $\square$

2. Any cut sequence

Each later cut acts inside one current piece, and Theorem 1 applies to that piece with its own s . The new hat is supported in the piece and vanishes at its ends; every earlier hat is linear there. The argument of § 1 gives orthogonality to all earlier hats and to the other pieces' remainders. After any finite set of cuts, with pieces τ_i ($\sum \tau_i = \tau$),

$$K_\tau = \sum_{\text{cuts}} \frac{m}{2} \int \dot{\phi}_{\text{cut}}^2 + \sum_i K_{\tau_i}, \quad \sum_i K_{\tau_i} = C \sum_i \tau_i^3 \leq C\tau \max_i \tau_i^2.$$

The remainder vanishes iff the mesh does ($\sum_i \tau_i^3 \geq (\max_i \tau_i)^3$). The positions and the order enter only through which pieces are cut, never through the total. $\square$

This is the Schauder (Lévy–Ciesielski) expansion of δx in the Cameron–Martin space $H_0^1[0, \tau]$, adapted to an arbitrary nested cut sequence; for dyadic cuts it is the textbook expansion (Ciesielski 1961; metadata). The reachability note has the same cubic law for the area $2F^2 t^3/(3m)$ of the reachable lens of a cut. The bridge's own conditional

variance shows the same nested additivity with a quadratic law: for diffusion constant D , $\int_0^L \text{Var}(X_u | \text{ends}) du = DL^2/6$, and a cut at fraction s removes $Ds(1-s)L^2/3$ of it.

3. The rod and the two Mohist schedules

Zhuangzi 33 (dialecticians' list, text): 一尺之捶，日取其半，萬世不竭， “a one-foot stick, each day take half, in ten thousand generations it is not exhausted.” Mohist Canon B (B60 in Graham' s numbering; canon): 非半弗斲，則不動，說在端， “without halving there is no cutting and no moving; the explanation lies in the point (端)” , with the explanation (text) 前則中無爲半，猶端也。前後取，則端中也 (working gloss: “going forward, at the middle nothing serves as a half: it is like a point; taking from front and back, the point is in the middle”), which continues 斲必半， “cutting must be by halves” .

One-ended schedule. Day k halves the remaining piece, of length $\tau 2^{-(k-1)}$, and spends $\frac{3}{4}C\tau^3 8^{-(k-1)}$. The total is $\frac{3}{4} \cdot \frac{8}{7} K_\tau = \frac{6}{7} K_\tau$; the pieces taken away keep $\sum_{k \geq 1} 8^{-k} K_\tau = \frac{1}{7} K_\tau$.

Alternating schedule (the reading favoured by 斲必半). One halving a day, taking the half alternately from the front and from the back. Each day is one halving of the remaining piece, so the spending is that of the one-ended schedule, $\frac{6}{7} K_\tau$; the remainder converges to the point $\frac{\tau}{2} + \frac{\tau}{8} + \dots = \frac{2\tau}{3}$.

Two quarter-cuts a day (a construction motivated by 前後取; the text does not fix the fractions). Day k cuts the remaining piece $r = \tau 2^{-(k-1)}$ at $s = \frac{1}{4}$ and $s = \frac{3}{4}$, keeping the middle half. The two cuts spend $C r^3 (1 - 2 \cdot \frac{1}{64} - \frac{1}{8}) = \frac{27}{32} C r^3$ (equivalently $\frac{9}{16} + \frac{9}{32}$ by two applications of Theorem 1), and the total is $\frac{27}{32} \cdot \frac{8}{7} K_\tau = \frac{27}{28} K_\tau$, with $\frac{1}{28} K_\tau$ kept in the pieces taken away. The contrast $\frac{6}{7}$ against $\frac{27}{28}$ is between one half-cut and two quarter-cuts a day, whichever ends are cut.

All three schedules converge to a point, the Mohist 端: the end, the point $2\tau/3$, and the midpoint. If 中 in 端中 means the midpoint, the explanation fits the symmetric schedule, while 斲必半 fits the alternating one. What each author is committed to, and what the theorem does with it:

- The dialecticians hold that the halving never ends. Theorem 2 agrees for the figure and adds a price: their schedule spends only $\frac{6}{7}$ of the action, since it never cuts again what it took away.
- The Mohists hold that the halving ends at a point. Theorem 2 agrees that each schedule converges to a point, and the point carries no action. The unhalvable 端 as a least magnitude is a reading the theorem leaves aside, as the Planck paper already records.
- Newton' s scholium closing Book I, Section I, takes the geometers' side (divisibility without end). Theorem 2 is that side made quantitative: refinement exhausts the action only when every piece is cut again.

The parallel with Zeno' s dichotomy (Aristotle, Physics VI.9) is recorded as convergence; transmission would need evidence this note does not have.

4. The weight of a cut

Take the Euclidean path measure of a free particle with variance rate $\hbar/m$: unconditioned centred Brownian motion X with $E[X_t^2] = \hbar t/m$, then pin it at the cell' s endpoints. Given the endpoints of a piece of length L , the value at fraction s is Gaussian with variance $\hbar s(1-s)L/m$, independent of the finer hats and of the other pieces, by the Markov property of the bridge (the Lévy construction, for any nested sequence). A

force F shifts the mean of that value by the sagitta $h = Fs(1-s)L^2/(2m)$. The Kullback–Leibler divergence of the shifted from the free law (the expected log-likelihood ratio, which here also equals the ratio evaluated at the shift) is

$$\frac{h^2}{2\hbar s(1-s)L/m} = \frac{mh^2}{2s(1-s)L\hbar} = \frac{3s(1-s)CL^3}{\hbar},$$

the spent share of Theorem 1 divided by $\hbar$. By the chain rule the divergences add over cuts, and along a sequence with mesh tending to zero they sum to $K_\tau/\hbar$, the Cameron–Martin exponent for the shift δx . The Cameron–Martin density factorizes over cuts, each contributing the normalizing factor $e^{-3s(1-s)CL^3/\hbar}$. This is the Euclidean counterpart of the real-time two-path phase K_τ/ε of the two-path note (the same number with ε in the role of $\hbar$), and of the product-over-cuts structure that § 7 of the fifth-postulate note reads in the time-step partition of Rivero 1998. For $\hbar \rightarrow 0$ every cut, however small, is decisive; for $\hbar > 0$ the cuts whose share falls below $\hbar$ carry little evidence (under one nat). $\square$

5. A floor counts the cuts

If every cut that a record exhibits must spend at least $\kappa > 0$, Theorem 2 gives at most K_τ/κ such cuts, for every schedule. For halving refinement the 2^n cuts of level n each spend $\frac{3}{4}K_\tau 8^{-n}$, so the exhibited levels are those with $8^n \leq 3K_\tau/(4\kappa)$.

Two floors must be kept apart: κ here floors the share a cut spends, $3s(1-s)K_L$, while the thresholds in the other notes floor the action of a cell, K_τ ; for halving the two differ by the factor $\frac{3}{4}$. The Planck paper’s eq. (2) is $K_\tau \geq 2z_{1-\epsilon}^2\hbar$: τ_* is the cell with $K_{\tau_*} = 2z_{1-\epsilon}^2\hbar$, whose halving spends $\frac{3}{2}z_{1-\epsilon}^2\hbar$. The round-3 resolution threshold $\tau \geq (24m\varepsilon d_{n,\eta}/F^2)^{1/3}$ is $K_\tau \geq \varepsilon d_{n,\eta}$, that is $\kappa = \frac{3}{4}\varepsilon d_{n,\eta}$ in cut-share form. Corollary 5 thus places a floor, when one is given, at a definite depth of the cut process; it supplies no floor.

5b. The mark mesh from the path measure (Proposition 6)

Keep the measure of § 4: unconditioned centred Brownian motion with $E[X_t^2] = \hbar t/m$, the Euclidean free-particle measure, whose diffusion coefficient $\hbar/2m$ is the one of Nelson (1966) (metadata). Assume $m, \tau, \hbar > 0$, $F \in \mathbb{R}$, equal priors and $0 < \epsilon < 1/2$. Let ℓ be a fixed endpoint chord and let B be the centred bridge with covariance $E[B_u B_v] = (\hbar/m)(\min(u, v) - uv/\tau)$. Compare $P_0 = \mathcal{L}(\ell + B)$ and $P_1 = \mathcal{L}(\ell - \delta x + B)$, where $\delta x(u) = Fu(\tau - u)/(2m)$. This specifies additive position noise around the two Newtonian comparison paths. Their Cameron–Martin norm is $d^2 = \|\delta x\|_{\text{CM}}^2 = (m/\hbar) \int \dot{\delta x}^2 = 2K_\tau/\hbar$. Writing $h = -\delta x$, the log-likelihood ratio is $Y - d^2/2$, where $Y = \langle h, X - \ell \rangle_{\text{CM}}$ denotes the Gaussian stochastic pairing, defined even though typical paths lie outside the Cameron–Martin space. Here integration by parts makes it the ordinary area statistic $Y = -(F/\hbar) \int_0^\tau (X_u - \ell_u) du$. Under P_0 , $Y \sim N(0, d^2)$; under P_1 , $Y \sim N(d^2, d^2)$. Thus the Neyman–Pearson test with equal priors thresholds that statistic halfway and errs with probability

$$p_{\text{err}} = \Phi\left(-\frac{1}{2}\|\delta x\|\right) = \Phi\left(-\sqrt{K_\tau/2\hbar}\right).$$

Hence $p_{\text{err}} \leq \epsilon$ iff $K_\tau \geq 2z_{1-\epsilon}^2\hbar$, that is $\tau\Delta E = 12K_\tau \geq 24z_{1-\epsilon}^2\hbar$ and, for $F \neq 0$, $\tau \geq \tau_* = (48z_{1-\epsilon}^2 m\hbar/F^2)^{1/3}$: eq. (2) of the Planck paper, including its “about $65\hbar$ at $\epsilon = 0.05$ ” ($2z_{0.95}^2 = 5.41$). A single cut observed alone gives $\Phi(-\sqrt{\text{share}/2\hbar})$ in the same way. $\square$

The Planck paper reaches (2) from real-time Gaussian marks whose record and recoil obey Robertson's bound $\hbar/2$; here the same constant comes from the Euclidean path measure $e^{-S/\hbar}$ through the Cameron–Martin theorem alone. The agreement is exact and structural. In Theorem 2 of the Planck paper, the mark trade-off $\kappa_{\text{mark}} = \hbar/2$ gives $\sup d_{\text{mark}}^2 = F^2 \tau^3 / (24m\kappa_{\text{mark}}) = 2K_\tau/\hbar$. Its optimization uses the same Dirichlet inverse of $-d^2/du^2$ with zero boundary values, hence the same parabolic extremizer and $\tau^3/12$. The full bridge attains this Gaussian information; the independent, uncorrelated, non-adaptive mark protocols approach it in the dense limit for tests invariant under initial position and velocity. This equality of sharp information retains the different experimental premises; it supplies no equivalence of all records or of Nelson's state-dependent dynamics. Both routes supply $\hbar > 0$, giving a second conditional derivation of the mesh, with the necessity question of the fifth-postulate note unchanged. What it adds is the reading of the floor on recording the inertial–parabola difference as the distinguishability threshold of the force in the free-particle path measure. At small ϵ , $z_{1-\epsilon}^2 \simeq 2\ln(1/\epsilon)$, so this floor grows like $4\hbar \ln(1/\epsilon)$ in cell action, the same logarithmic law as Theorem E of the fifth-postulate note, $ab \geq \hbar c_*(\epsilon)$ with $c_*(\epsilon) \simeq \frac{1}{2} \ln(1/\epsilon)$ in window half-widths; the constants refer to different quantities (a cell's Galileo action and a phase-space window).

5c. Mark protocols spend exactly the cut measure (Proposition 7)

Proposition 7 (2026-09-29, Claude, generalizing a single-cut computation by Fable; written proof, unrefereed). Take the mark model of the Planck paper, § 3: mark j returns the position with Gaussian error δ_j and delivers a Gaussian impulse Δ_j with $\delta_j \Delta_j \geq \kappa$; protocols are non-adaptive and tests invariant under the unknown initial position and velocity. Let the marks sit at $t_0 < t_1 < \dots < t_N$, with $\tau = t_N - t_0$ and pieces $\tau_i = t_i - t_{i-1}$. Then the supremum over the marks' δ_j, Δ_j of the discernibility is

$$\sup d^2 = \frac{K_\tau - \sum_i K_{\tau_i}}{\kappa} = \frac{1}{\kappa} \sum_{\text{cuts}} \frac{m}{2} \int \dot{\phi}_{\text{cut}}^2, \quad K_\tau = \frac{F^2 \tau^3}{24m},$$

the action that the interior cuts spend by Theorems 1–2. It is approached with sharp end marks and balanced interior marks. Special cases: a single cut at fraction s gives $3s(1-s)K_\tau/\kappa$; N equal pieces give $(1 - N^{-2})K_\tau/\kappa$, so n halvings give $(1 - 4^{-n})K_\tau/\kappa$; the dense limit gives K_τ/κ , the paper's Theorem 2. Adding a mark inside a piece adds exactly that cut's share, $3s(1-s)K_{\tau_i}/\kappa$: the squared statistical distance is additive under refinement at every finite stage.

Proof. Write $T(w) = \sum_{t_i > w} u_i(t_i - w)$ for the test weights u . The invariance constraints $\sum u_i = \sum u_i t_i = 0$ say exactly that T is piecewise linear with kinks at the t_j and vanishes outside (t_0, t_N) , and every such T arises from one u , with u_j the jump of T' at t_j . The paper's § 3 gives the signal $\frac{F}{m} \int T$ and the noise $\sum_j \delta_j^2 u_j^2 + \sum_j \Delta_j^2 T(t_j)^2/m^2 \geq \frac{2\kappa}{m} \sum_j |u_j| |T(t_j)|$, by the arithmetic–geometric mean. Summation by parts gives $\sum_j u_j T(t_j) = -\int T'^2$, hence

$$d^2 \leq \frac{F^2}{2m\kappa} \frac{(\int T)^2}{\int T'^2}.$$

Let V_h be the piecewise-linear functions on the grid vanishing at t_0, t_N , and $T_h \in V_h$ the Galerkin solution of $-T'' = 1$: $\int T_h' \varphi' = \int \varphi$ for all $\varphi \in V_h$. Cauchy–Schwarz in the energy product gives $(\int T)^2 \leq \int T_h'^2 \int T'^2$ with $\int T_h'^2 = \int T_h$, so the ratio is at most $\int T_h$. In one dimension T_h interpolates $p(w) = \frac{1}{2}(w - t_0)(t_N - w)$ at the nodes, since the Dirichlet Green's function with its source at a node lies in V_h and the error $p - T_h$ is Galerkin-orthogonal to V_h . So $\int T_h$ is the trapezoid rule for p , whose error on a piece is $-\tau_i^3 p''/12 =$

$\tau_i^3/12$, and $\int T_h = (\tau^3 - \sum_i \tau_i^3)/12$. This gives the upper bound $F^2(\tau^3 - \sum_i \tau_i^3)/(24m\kappa)$. For attainment take $T = T_h$, which is nonnegative and concave, so all $u_j T(t_j)$ have one sign; choose $\delta_j = \Delta_j T_h(t_j)/(m|u_j|)$ at interior marks, and let the end marks be sharp, $\delta_0, \delta_N \rightarrow 0$ with $\Delta_0, \Delta_N \rightarrow \infty$, which costs nothing because T vanishes there and their impulses reach no later reading. $\square$

The two constructions of this note and the paper' s are therefore one: the Newton cell action spent by any finite set of cuts is, over κ , the best discernibility of a mark protocol placed at those cuts. With $\kappa = \hbar/2$ the dense value reproduces Proposition 6' s exponent, $\frac{1}{2}\sqrt{K_\tau/\kappa} = \sqrt{K_\tau/(2\hbar)}$, from the mark model instead of the path measure. The continuity note reads the additivity as Leibniz' s petites perceptions.

6. Consequence for STATE

Atlas § 1b: the Newton row of “cuts at any position” is proved here (Theorems 1–2, Corollary 3), with the Lévy–Ciesielski identification and the Euclidean weight of a cut (Proposition 4). Proposition 6 is now refereed: its Gaussian information equals the sharp mark information under the respective stated hypotheses. The necessity question is unchanged. On the gauge side of § 1b the free-field defect of a cut at fraction s is (5) scaled by $4s(1-s)$ in its size bound (Corollary 2_s); the $U(1)$ Theorem 5 holds for every cut fraction (Corollary 5_s there).